

Mean-Reverting Logarithmic Models of the VIX, Revisited:

A Reproduction of Bao (2013) and an Out-of-Sample Test on the 2026 VIX Option Chain

Julian Philip Henry

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Abstract

Bao (2013) models the logarithm of the VIX as a mean-reverting process, with and without upward jumps (MRLRJ) and stochastic vol-of-vol (MRLRSV), and calibrates the models to VIX options from September 2011. Neither the quotes nor the calibration protocol were published. We give an open implementation, verify it against the thesis’ published parameter tables, and re-run the thesis’ calibration procedure unchanged on the Cboe VIX option chain of 29 September 2026. The central qualitative claims survive: the pure-diffusion MRLR model cannot produce the VIX skew (pricing errors 28–38%), while jumps *or* stochastic vol-of-vol each fit it (errors 1.6–2.8%, 92–100% of model prices inside the bid–ask spread, and similar errors on held-out strikes). Three further findings qualify them. First, the fitted parameters are largely not identified: in MRLR the mean-reversion speed has no effect on the fit at all, and the jump model’s per-maturity fits drift to $\kappa \approx 0.01$, $\theta \approx -20$. Second, the 2026 smile needs rare, large jumps ($\lambda \approx 2$ –5/yr, mean log-jump 0.3) or vol-of-vol three to seven times the 2011 estimates, far from the 2011 regime of frequent small jumps. Third, no model fits the four maturities with one set of parameters: the best (MRLRJ) prices only half the quotes inside the spread. The models work as per-maturity smile interpolators, not as a consistent description of VIX dynamics. Code, data snapshot and all fitted parameters are released.

1 Introduction

Options on the VIX are among the most liquid volatility derivatives, yet their underlying is unusual: the index is not traded, it reverts quickly toward a long-run level, and it jumps up in market stress far more readily than it falls. VIX options therefore trade with a *positive* implied-volatility skew, the reverse of equity index options, and VIX futures carry a term structure that is usually in contango and flips to backwardation in crises.

Bao [1] builds on the direct-VIX approach of Whaley [2], Grünbichler and Longstaff [3] and Psychoyios et al. [4]: instead of deriving the VIX from a model of the S&P 500, it specifies dynamics for $\ln \text{VIX}_t$ directly and prices futures and options in closed form or through the characteristic function. The thesis studies four nested specifications, derives futures, options, hedging ratios and convexity adjustments for each, and ends with an empirical comparison on Cboe quotes from 26 September 2011 (spot VIX 42.3). Its conclusions are that the

log-normal MRLR model cannot generate the skew, that adding jumps *or* stochastic vol-of-vol is enough, and that combining both adds nothing.

The empirical chapter cannot be reproduced as published: the option quotes, the optimizer, the starting values and the parameter bounds are not reported. This note (i) releases an open Julia implementation of MRLR, MRLRJ and MRLRSV; (ii) tests it against the parameter tables published in the thesis; (iii) applies the thesis’ calibration procedure, unchanged, to the Cboe VIX chain of 29 September 2026 (spot VIX 16.04); and (iv) asks four questions the original study did not: do the fits hold on held-out strikes, are they consistent with VIX futures, are the parameters identified, and can one set of parameters explain the whole term structure?

2 The models and why each exists

Why logarithms, why mean reversion. A model of the VIX must keep it positive, pull it back toward a long-run level after spikes, and let its volatility scale with its level. The mean-reverting square-root (MRSR) process of Grünbichler and Longstaff [3] satisfies the first two, but its volatility grows only with $\sqrt{\text{VIX}}$ and it has a thin right tail. Modelling $X_t = \ln \text{VIX}_t$ as an Ornstein–Uhlenbeck process instead makes the VIX log-normal with level-proportional volatility:

$$dX_t = \kappa(\theta - X_t) dt + \sigma dW_t. \quad (\text{MRLR})$$

Here κ is the speed of mean reversion, e^θ the long-run VIX level and σ the vol-of-vol. With $\phi = e^{-\kappa\tau}$, X_T is Gaussian with mean $\mu = \phi X_t + (1 - \phi)\theta$ and variance $v = \sigma^2(1 - \phi^2)/(2\kappa)$, so the VIX future is $F = e^{\mu+v/2}$ and a call is a Black formula on F . The thesis lets θ_t and σ_t depend on time so that the model reproduces the futures curve and the at-the-money volatility term structure by construction.

Why jumps. Because MRLR is log-normal, its implied volatility against the future is the same at every strike: it cannot produce a skew. The VIX’s defining feature is sudden upward jumps, so the thesis (following Psychoyios et al. [4]) adds compound-Poisson jumps with exponential sizes:

$$dX_t = \kappa(\theta - X_t) dt + \sigma dW_t + J dN_t, \quad (\text{MRLRJ})$$

with $N_t \sim \text{Poisson}(\lambda)$ and $J \sim \text{Exp}(\eta)$, i.e. mean log-jump $1/\eta$. Upward jumps fatten the right tail and create

a positive skew, most strongly at short maturities before mean reversion dampens them. The characteristic function $\psi(s) = \mathbb{E}_t^{\mathbb{Q}}[e^{isX_T}]$ stays in closed form:

$$\ln \psi(s) = is\mu - \frac{1}{2}s^2v + \frac{\lambda}{\kappa} \ln \frac{\eta - is\phi}{\eta - is}. \quad (1)$$

The jumps are not compensated, so $F = \psi(-i)$ is finite only if $\eta > 1$.

Why stochastic vol-of-vol. Vol-of-vol itself varies: it rises when the VIX rises. MRLRSV replaces the constant σ by a square-root variance process correlated with the VIX shocks:

$$\begin{aligned} dX_t &= \kappa(\theta - X_t) dt + \sqrt{V_t} dW_t, \\ dV_t &= \kappa_v(\theta_v - V_t) dt + \sigma_v \sqrt{V_t} dZ_t, \end{aligned} \quad (\text{MRLRSV})$$

with $d\langle W, Z \rangle = \rho dt$. A positive ρ means that VIX rallies come with higher vol-of-vol, which fattens the right tail as jumps do but builds up gradually, so its effect grows with maturity. The model is affine: $\psi = \exp(a + bV_t + is\phi X_t)$, where, with $u(h) = is e^{-\kappa h}$,

$$\begin{aligned} b' &= \frac{1}{2}u^2 + \rho\sigma_v u b + \frac{1}{2}\sigma_v^2 b^2 - \kappa_v b, \\ a &= is\theta(1 - \phi) + \kappa_v \theta_v \int_0^T b dh, \end{aligned} \quad (2)$$

$b(0) = 0$. The Riccati equation has time-dependent coefficients; its Kummer-function solution is unstable, so the thesis, and we, solve it by Runge–Kutta.

Why not both. MRLRSVJ combines jumps and stochastic vol-of-vol. In the thesis it fits no better than MRLRSV (percentage errors within 0.01 points) at the cost of two more parameters, so we do not implement it.

3 Pricing and verification

Calls follow the Heston-type formula of the thesis (Theorem 4.3), $C = D(F\Pi_1 - K\Pi_2)$, with the Gil-Pelaez [5] probabilities

$$\Pi_2 = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\text{Im}(e^{-iuk}\psi(u))}{u} du, \quad k = \ln K,$$

and Π_1 the same with $\psi(u - i)/\psi(-i)$. The integrals use composite 16-point Gauss–Legendre quadrature on $[0, U]$, with U grown until $|\psi(U)| < 10^{-12}$ and one grid shared by all strikes of a maturity. For MRLRSV, (2) is solved by RK4 at each node, with a step that resolves $e^{-\kappa h}$ and keeps $h(\sigma_v|s| + \kappa_v) \leq 1$ inside the stability region of this stiff equation.

A test suite (34 assertions) checks that the Fourier pricer reproduces the MRLR closed form to 10^{-8} ; that MRLRJ ($\lambda \rightarrow 0$) and MRLRSV ($\sigma_v \rightarrow 0$, $V_0 = \theta_v$) reduce to MRLR; that $F = \psi(-i)$ matches the closed-form futures; that prices are decreasing, convex and within

Table 1: Published 2011 parameters priced with our implementation: mean absolute price difference between models (strikes 20–90) and the thesis’ reported MAE against the market (Table 7.3).

Expiry	J – SV	MAE _J +MAE _{SV}	LR – J	MAE _{LR}
18 Oct 2011	0.035	0.238	0.173	0.433
15 Nov 2011	0.019	0.359	0.105	0.329
20 Dec 2011	0.022	0.409	0.079	0.357
17 Jan 2012	0.022	0.261	0.121	0.357

arbitrage bounds; that Euler Monte Carlo prices (200k paths) agree within four standard errors for MRLRJ and MRLRSV; and that a tenfold smaller RK4 step changes prices by less than 10^{-4} . A 25-digit BigFloat quadrature agrees with the double-precision pricer to 10^{-9} , so extended precision is unnecessary.

4 Consistency with the published tables

The 2011 quotes are not public, so the thesis’ errors cannot be recomputed. Its parameter tables can still be tested. Tables 7.5 (MRLRJ) and 7.6 (MRLRSV) were fitted separately to the *same* quotes, with reported mean absolute errors (MAE) of 0.12–0.21 each. Priced with our code ($VIX_t = 42.3$, strikes 20–90), the two structurally different models agree to 2–3.5 cents at every maturity (Table 1), well inside that tolerance; MRLR (Table 7.4) differs by 8–17 cents. Against the model future, the MRLR implied volatility is flat and the MRLRJ and MRLRSV curves slope upward, at the levels plotted in the thesis’ Figure 7.3 (e.g. 1.27/1.39/1.48 for MRLRJ at $K = 40/60/80$ in October). A convention error in the jump or Riccati terms would break this agreement, so the implementation matches the thesis.

The tables also imply numbers the thesis never reports: model VIX futures of 37.8–38.5 (Oct-2011), 35.0–35.4 (Nov), 31.7–32.6 (Dec) and 32.2–33.4 (Jan-2012), a backwardated curve consistent with the text. They also show that the calibration ran into undisclosed bounds: $\theta = 3.00$ in all four MRLRJ fits and $\rho = 1.00$ in all four MRLRSV fits are boundary values, not estimates.

5 Data and calibration protocol

Data. We use the Cboe delayed-quote snapshot of the full VIX option chain at the close of 29 September 2026 (spot VIX 16.04 at 16:15 ET; 1,520 contracts; SHA-256 c404ecd5...ae5f9) [6]. As in the thesis we take the first four monthly maturities (21 Oct, 18 Nov, 16 Dec 2026 and 20 Jan 2027; open interest falls sharply after January), keep calls with positive bid *and* positive open interest (55, 63, 58 and 59 quotes) and fit mid quotes. Time to expiry runs to the 09:30 ET settlement on the expiry date (ACT/365).

Futures and rate. VIX options settle on the same quotation as VIX futures, so put–call parity holds against the future: $C - P = e^{-r\tau}(F - K)$. Fitting this jointly over all call–put pairs gives $r = 4.46\%$ and futures of 17.59, 18.28, 18.67 and 19.41, a contango curve above the spot. The thesis instead used spot VIX with no discounting adjustment.

Loss and fit measures. Exactly as in the thesis (Eq. 7.3): $\mathcal{L} = \sum_j (C_j^{\text{mod}} - C_j^{\text{mkt}})^2 + \alpha \sum_j (\ln C_j^{\text{mod}} - \ln C_j^{\text{mkt}})^2$ with $\alpha = 8$, which balances in- and out-of-the-money quotes. We report the percentage error $\text{PE} = \text{mean}|C^{\text{mod}} - C^{\text{mkt}}|/C^{\text{mkt}}$, the MAE and, as a stricter test, the share of model prices inside the bid–ask band.

Optimisation. This is the part the thesis omits. Parameters are mapped to unconstrained coordinates: log for positive parameters, $\eta = 1 + e^x$, $\rho = \tanh x$. MRLRSV uses logistic box bounds $\kappa \in [0.1, 50.1]$, $\theta \in [1, 5]$, $\kappa_v, \theta_v \in (0, 50)$ and $\sigma_v, V_0 \in (0, 30)$, which rule out a degenerate ridge ($\kappa \rightarrow 0$, $\theta \rightarrow -\infty$) and extreme stiffness. Our first bounds, $\sigma_v, V_0 \leq 5$, were binding on the October slice and were widened, since a 2026 short-dated smile needs far more vol-of-vol than in 2011. Each fit runs Nelder–Mead from 10 random starts (6 for MRLRSV), with seeds $1 + k$ and log-uniform ranges, followed by one restart from the optimum; each run is capped at 3,000 iterations and 300 s. The “paper procedure” fits every parameter, including θ , per maturity with spot VIX as input, as in the thesis.

6 Results on the 2026 chain

Fit quality. Table 2 repeats the thesis’ Tables 7.2–7.3 on the 2026 data. The ranking is the same and the gaps are much larger. MRLR misses by 28–38% and prices only about a fifth of the quotes inside the spread. MRLRJ and MRLRSV are essentially tied, with errors of 1.6–2.8% (lower than the thesis’ 3.0–5.2% in 2011) and 92–100% of prices inside the spread. As in 2011, the difference between jump and stochastic-vol models is within noise. Figure 1 shows why: the 2026 skew is steep, rising from about 0.7 in implied volatility at the money to about 2.0 at $K = 50$ for the October expiry. MRLR’s smile is flat against its *own* future, so it can match either the level or the slope of this curve but not both.

Held-out strikes. Refitting on strikes of odd rank and pricing the rest (Table 3) leaves MRLRJ almost unchanged (1.9–2.6% except 4.7% in January). MRLRSV degrades a little more (2.2–3.5%). The fits interpolate the smile; they are not memorising individual quotes.

Table 2: Paper procedure on 29 Sep 2026: percentage error, MAE and share of model prices inside the bid–ask spread. Thesis (2011) PE: MRLR 4.7–6.9%, MRLRJ 3.1–5.2%, MRLRSV 3.0–5.1%.

Expiry	PE (%)			MAE			in spread (%)		
	LR	J	SV	LR	J	SV	LR	J	SV
Oct-26	37.9	1.9	1.6	.527	.007	.008	18	100	100
Nov-26	29.5	2.7	2.8	.517	.016	.019	21	98	98
Dec-26	30.6	2.3	2.4	.516	.014	.017	22	98	98
Jan-27	28.3	2.4	2.4	.527	.020	.021	22	93	92

Table 3: Out-of-sample strikes: fit on odd-ranked strikes, evaluate on the rest. PE in %; last columns: share inside the spread.

Expiry	out-of-sample PE			in spread (%)	
	MRLR	MRLRJ	MRLRSV	J	SV
Oct-26	39.6	1.9	3.5	100	96
Nov-26	30.2	2.6	2.6	100	94
Dec-26	31.9	2.2	2.2	100	93
Jan-27	29.3	4.7	3.2	90	90

Futures consistency. The thesis’ procedure uses spot VIX as the underlying and never looks at futures, which the thesis itself flags as problematic (Sec. 7.3.2). In 2026 this matters only for MRLR. The MRLRJ and MRLRSV fits reproduce the parity-implied futures to within 0.1% although the futures were never targeted. Deep in-the-money calls pin the forward, and the free per-maturity θ absorbs the gap between spot and future. The MRLR fits put the future 27–30% below the market (12.8 vs. 17.6 in October): to fit the right tail through the log-loss, the log-normal model gives up the level.

Identification. The parameters are far less meaningful than the fit quality suggests. In MRLR, one maturity determines only the mean and variance of X_T , so (κ, θ, σ) has one redundant direction. Fixing κ at 0.5, 2, 8 or 32 and refitting θ, σ gives the same loss to nine digits (120.910550). The thesis’ MRLR values $\kappa = 11.05, 11.43, 9.88$ and 12.58 are therefore arbitrary points on a flat ridge. The same ridge appears in the 2026 MRLRJ fits: beyond October the optimum drifts to $\kappa \approx 0.01$ and $\theta \approx -15$ to -23 (Table 5). Only the product $\kappa\theta \approx -0.21$ to -0.26 is pinned down, so the jump model degenerates into a drifting Brownian motion with jumps, with no detectable mean reversion within any single maturity. In MRLRSV, κ_v and θ_v collapse toward zero at three maturities, so the variance process effectively has no mean reversion either, and in 2011 θ and ρ sat on bounds. Per-maturity MRLR-family parameters should not be read as estimates of VIX dynamics.

A different volatility regime. The parameters that are identified have moved a long way since 2011. There, MRLRJ used many small jumps ($\lambda = 60$ –170/yr, mean log-jump $1/\eta \approx 0.10$ –0.15), close to extra diffusion. In

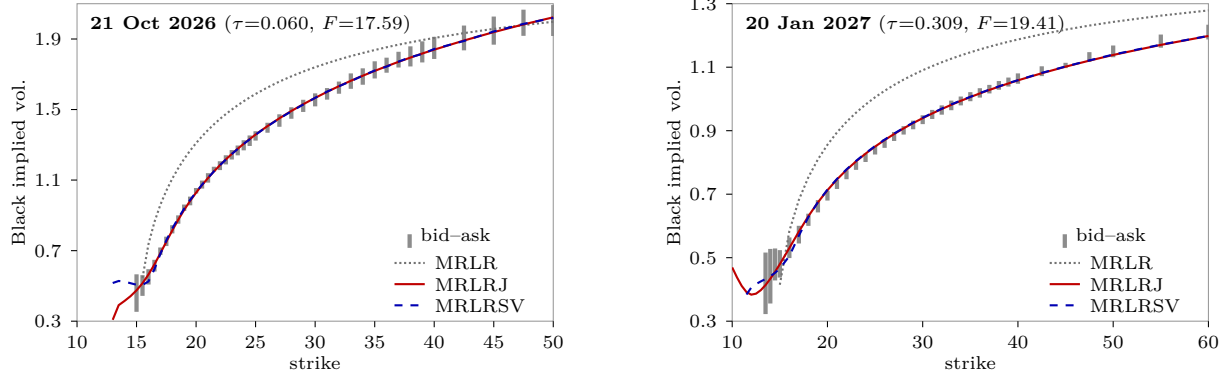


Figure 1: Black implied volatility against the parity-implied VIX future for the nearest and farthest calibrated maturities, 29 Sep 2026. Grey bars: market bid-ask; lines: models calibrated per maturity with the thesis’ procedure. MRLR is flat against its own future, which is 27–30% below the market (Sec. 6), so against the market future its curve is misplaced; MRLRJ and MRLRSV follow the steep positive skew.

Table 4: One parameter set for all four maturities (θ anchored to each future). PE in %. *time-capped optimisation.

Model	Oct	Nov	Dec	Jan	in spread
MRLR	95.0	81.0	71.5	68.7	9%
MRLRJ	12.6	6.8	5.3	6.3	52%
MRLRSV*	16.1	12.1	29.0	31.1	35%

MRLRJ: $\kappa = 1.20$, $\sigma = 0.469$, $\lambda = 2.33$, $\eta = 3.12$; MRLRSV: $\kappa = 1.80$, $\rho = 0.96$, $\kappa_v = 18.1$, $\theta_v = 0.18$, $\sigma_v = 8.17$, $V_0 = 1.59$.

2026 the October fit uses rare large jumps: $\lambda = 5.2/\text{yr}$, mean log-jump 0.30 (a 35% move in the VIX), about 0.3 jumps expected before expiry, on top of a small diffusion $\sigma = 0.35$. Likewise MRLRSV needs $\sigma_v = 3.2$ –14.5 against 0.57–1.98 in 2011. At a spot of 16, far below the 42 of 2011, the market prices a calm VIX with a large, discrete crash-spike risk.

One model for the term structure. Finally, we fit a *single* parameter set to all four maturities, with θ set per maturity so that each model future matches the market exactly. This is the thesis’ own two-stage strategy (Theorems 3.3, 4.4, 5.4) with constant vol-of-vol. Table 4 shows that the per-maturity success does not carry over. MRLRJ, the best, misses by 5–13% and prices only 52% of quotes inside the spread; MRLRSV does worse (12–31%); MRLR fails (69–95%). The MRLRSV run hit its time limit, so its figures are an upper bound, but the gap to the per-maturity fits is far larger than any plausible optimizer slack. The thesis anticipates this by making σ_t time-dependent, which effectively reintroduces per-maturity parameters.

7 Discussion and limitations

The results rest on a single end-of-day snapshot of delayed quotes; a panel of dates would show how stable the regime contrast with 2011 is. Futures are inferred from put–call parity rather than taken from traded VIX futures, although the parity fit is tight. Nelder–Mead

with random restarts is a local method: for the per-maturity fits, different seeds reproduced the errors to two decimals, but the joint MRLRSV fit is not converged. MRLRSVJ and the time-dependent σ_t of the thesis are not implemented. The loss is the thesis’ MMLSE; an implied-volatility loss would weight the wings differently.

8 Conclusion

The main empirical claims of Bao (2013) hold up in a very different market, fifteen years later. Log-normal mean reversion alone cannot price VIX options, and upward jumps or stochastic vol-of-vol, not both, are enough to fit a single maturity to within the bid-ask spread. What the thesis left out matters as much. The fitted parameters sit on flat ridges and bounds, the 2026 smile implies rare large jumps rather than the frequent small ones of 2011, and no constant-parameter member of the family explains the term structure. For pricing a given expiry, MRLRJ is a compact and accurate smile model; as a model of how the VIX moves, the family needs maturity-dependent parameters or richer dynamics.

Code and data. The Julia package, tests, the 2026 snapshot and the scripts that regenerate every table and figure are in `julia_impl/`: `Pkg.test()`, `scripts/reproduce_paper.jl`, `scripts/calibrate_2026.jl` and `scripts/make_figures.jl`.

References

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Table 5: Full disclosure: all parameters from the thesis’ per-maturity procedure on 29 Sep 2026 (spot VIX 16.04, $r = 4.46\%$). Values at or near a bound are marked † . Per-strike fits: `results/calibration_2026-09-29_curves.csv`.

Expiry	MRLR			MRLRJ					MRLRSV						
	κ	θ	σ	κ	θ	σ	λ	η	κ	θ	ρ	κ_v	θ_v	σ_v	V_0
Oct-26	88.5	2.35	8.35	9.29	2.76	0.353	5.23	3.29	29.1	2.86	0.93	1.31	13.6	14.5	1.74
Nov-26	19.6	2.17	5.15	0.015	-15.0	0.263	2.93	3.45	3.48	2.98	0.75	0.20	$0^\dagger$	5.39	0.92
Dec-26	0.265	-8.15	1.99	0.014	-18.7	0.265	2.38	3.42	2.07	3.00	0.71	$0^\dagger$	$0^\dagger$	3.48	0.75
Jan-27	0.325	-2.68	1.69	0.009	-23.3	0.202	2.11	3.54	7.13	2.90	0.99	0.16	$0^\dagger$	3.17	1.77

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